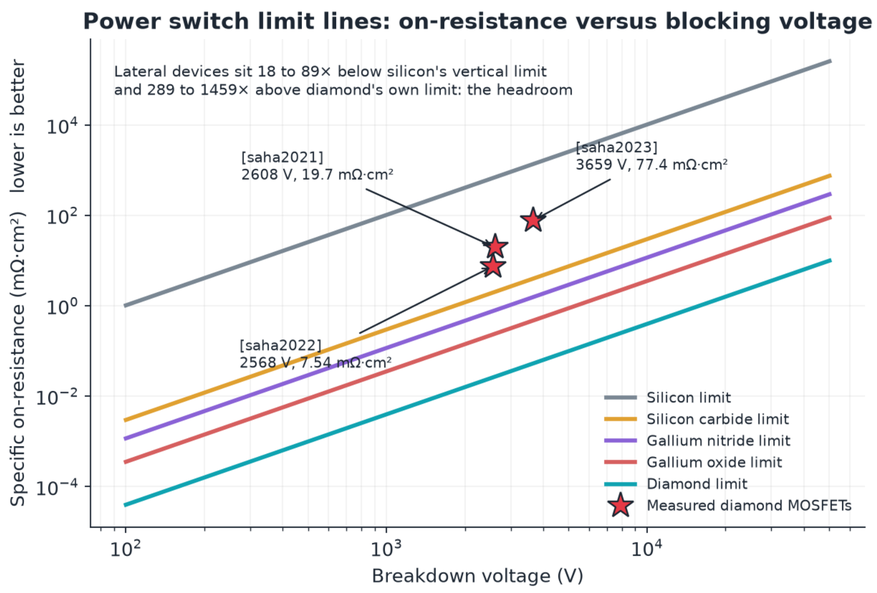
Diamond chips, from wafer to qubit

Aleksander Norman · California State University, Dominguez Hills · github.com/Normansrule/adamas-diamond-framework

ADAMAS: an open, tested framework for diamond electronics and room-temperature quantum processors

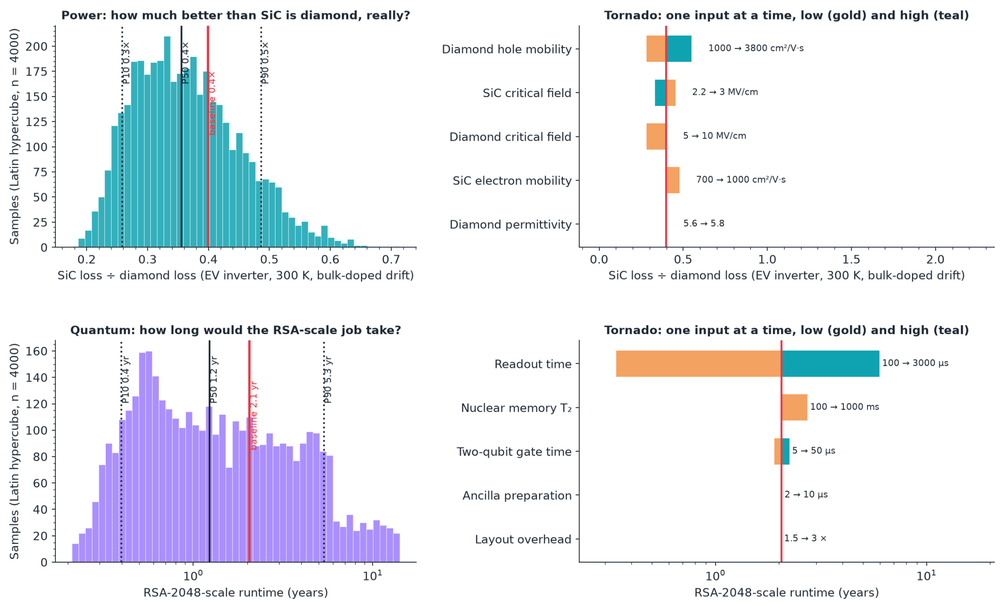
Diamond has the largest breakdown field and thermal conductivity of any practical semiconductor, and a defect, the nitrogen-vacancy (NV) center, that is a qubit at room temperature. ADAMAS follows one path from wafer to quantum processor with executable, cited models: **583 references**, **53 regenerable figures**, a process traveler, a placed processor, circuit-level error correction, and experiments from US$100. **Every number on this poster is generated by the package.**

1 · Power: diamond wins when hot



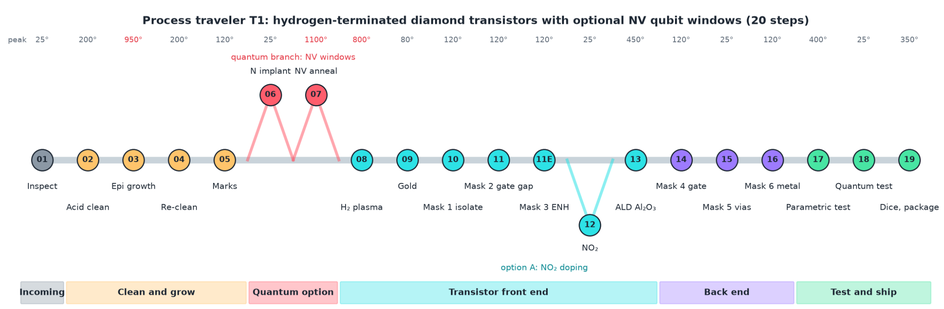
Sources: [baliga1982] [donato2020] [saha2021] [saha2022] [saha2023] [kasu2022talk]

Measured lateral hydrogen-terminated diamond MOSFETs reach 3659 V and sit 89× below silicon's vertical limit. Only 0.50% of the boron in a bulk-doped 1.2 kV drift is ionized at 300 K, so in an 800 V electric-vehicle inverter a bulk-doped diamond switch loses 74.6 W against 29.7 W for silicon carbide (ratio 0.40), and wins when hot: **1.9×** at a 523 K junction (median 1.7× with material uncertainty). Sobol indices rank what to measure first: diamond hole mobility, by a Hall measurement costing a few thousand dollars.



Sources: [isberg2002] [wort2008] [donato2020] [kimoto2014] [hopper2018] [maurer2012] [dolde2013] [litinski2019]; adamas.uncertainty (ranges are this repository's judgment)

2 · How to build it



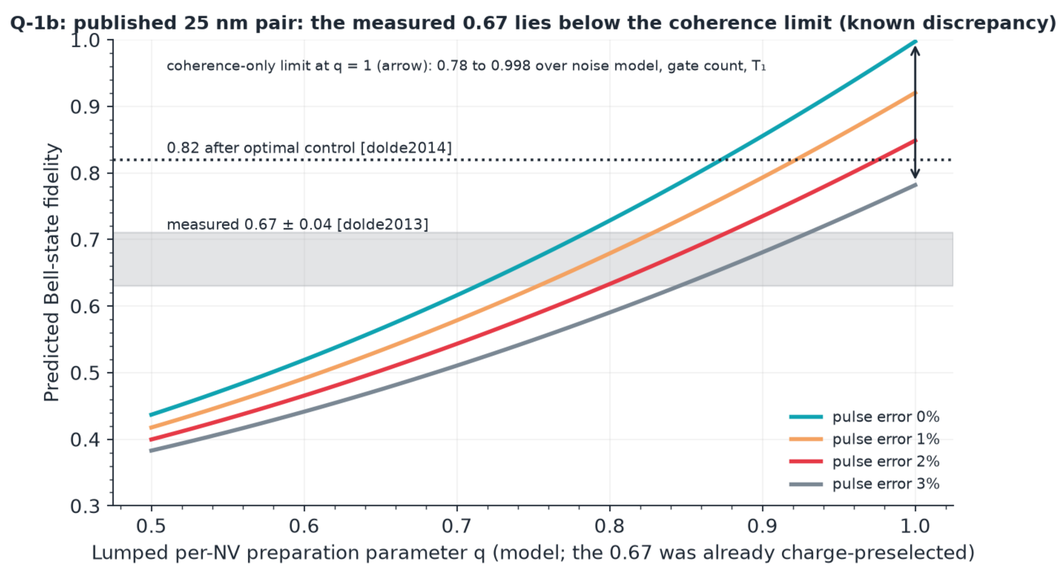
Sources: [kawarada2023] [kasu2012] [pezzagna2010] [hauf2011] [lu2013]; adamas.traveler

A 20-step traveler: hydrogen termination makes the channel, gold protects it, one oxygen plasma isolates devices, and all heat is spent before the first metal. Wafers that already hold NV centers need the low-temperature ($\leq$500 °C) hydrogen option, because a hot hydrogen plasma passivates them. The PDK-0 kit places the 4-bit DIA-4 processor (323 cells) in about 1 mm^2.

Validation and reproducibility

18 validation checks (17 pass) against published results, independent tools, and known answers; 500 references verified

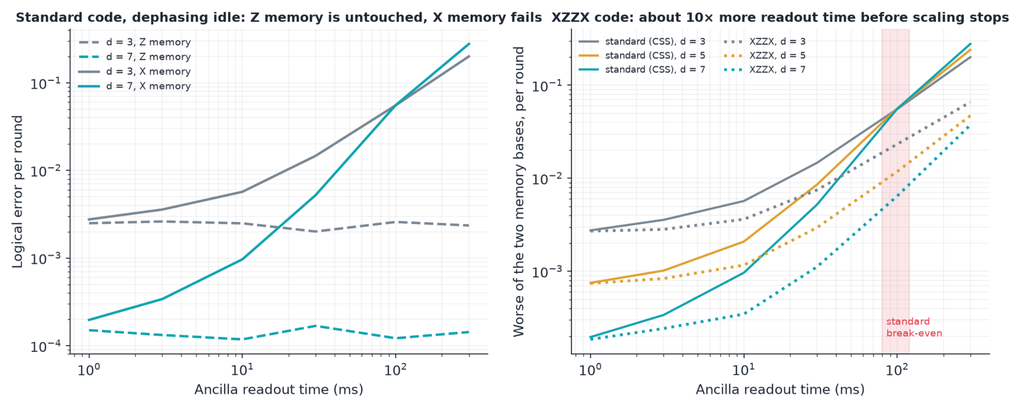
3 · Two-qubit gates: an open discrepancy



Sources: [dolde2013] [dolde2014]; adamas.gate_budget model (cubic echo exponent, one gate application for the curves)

With the published 25 nm pair, coherence alone allows a Bell fidelity of **0.78 to 0.998**, depending on the noise model. The measured 0.67 (charge-preselected) lies below that whole range: a known discrepancy. The earlier preparation explanation is withdrawn after the audit.

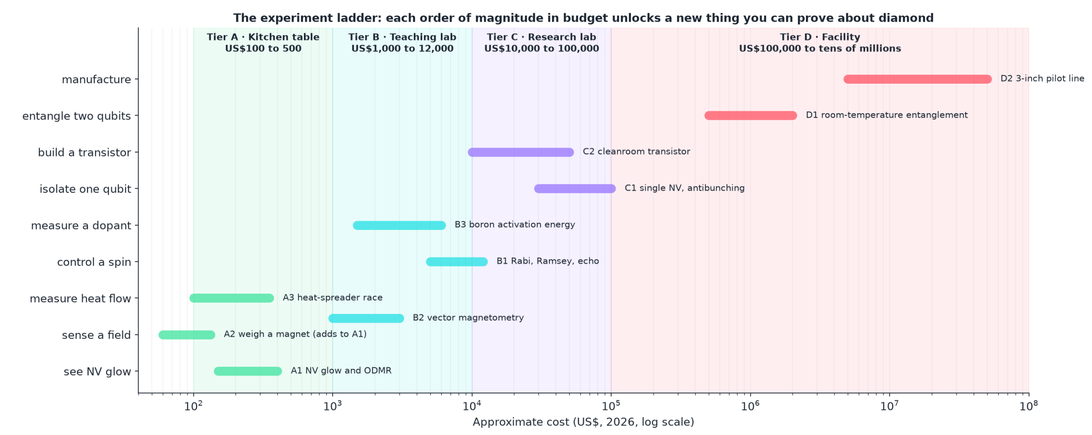
4 · Error correction with slow readout



Sources: [tuckett2018] [bonillaataides2021] [gidney2021stim] [higgott2022] [maurer2012]; adamas.circuit_qec (XZZX emulated by Hadamard conjugation of a checkerboard sublattice)

Circuit-level simulations (Stim, PyMatching): scaling stops near 100 ms readout for a 1 s nuclear memory; the bias-tailored XZZX code extends this to about **1000 ms**. With two-level qubits only CZ keeps a bias (CNOT at most 5): native XZZX circuits then reach a threshold of **1.81%** at bias 100, against 0.86% for the standard code. RSA-2048, calibrated to the published 8-hour estimate, fits on a 5.5 mm die but runs **2.1 years** at 1 ms readout (P10 to P90: 0.4 to 5.3), assuming a 25 μs, 0.3% inter-cell link that does not exist yet. With biased gate noise the standard code needs 173 M qubits; XZZX needs **58 M**.

5 · Try it: experiments from US$100



Sources: adamas.experiments (prices are approximate street prices; verify with vendors)

Ten experiments in four budgets with parts, steps, safety, and expected results; the A1 ODMR kit ships with Raspberry Pi Pico firmware and a fitter.

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